GANDHARV PATIL

701 C Sue Holloway Drive, Nepean, Ontario
+1 514-571-8714 ◇ ✉ ◇ ◇ ◇ Google Scholar

RESEARCH INTEREST

Decision Making at scale, Reinforcement Learning, Generative Models and Probabilistic Inference.

EDUCATION

McGill University/ Mila, Québec AI Institute
Degree(s): PhD, *Sept 2023- Present*
Masters in Engineering *Jan 2017 - Dec 2019*
GPA: 3.92/4.0
Supervisor(s): **Doina Precup** , **Pablo Samuel Castro**
Awards & Honours: McGill Graduate Excellency Fellowship (2025, 2017-2019), Google research award- (2018-2019), Deepmind research award - (2019-2020).

University of Puné *Sep 2009 - Jun 2013*
Degree: Bachelor of Engineering - Electronics & Telecommunication
Thesis: Activity Monitoring using Optical flow estimation and Kalman filters
Grade: First class with Distinction, GPA: 3.7/4.0 [1]

RESEARCH & INDUSTRIAL EXPERIENCE

ServiceNow Research *April 2026 - Present*
Visiting Researcher

- Developing methods for continual learning and long horizon RL in LLMs for enterprize tasks.

Amazon *Sep 2025 - Dec 2025*
Applied Scientist Intern (Market Science Team)

- Developed a multi-agent reinforcement learning method to improve performance and adaptability of Amazon freight's market place algorithms. Implemented a large scale agent-based discrete time simulation engine for evaluation and testing of RL methods.

RBC Borealis *Feb 2025 - Aug 2025*
Machine Learning Researcher Intern

- Developed a novel causal inference framework for estimating individual treatment effects (ITE) and treatment effect heterogeneity across subpopulations, under unobserved confounding.
- Developed a framework based on Epistemic Neural Networks for efficient and scalable uncertainty quantification for individual units particularly for unseen or out-of-distribution cases [1].

Google Deepmind *Nov 2024 - Jan 2024*
Student Researcher

- Developing methods to enhance reasoning and planning in LLMs (Gemini) to integrate external tools for domain-specific information retrieval and dynamic multi-step task execution.
- Improving the state-of-the-art LLM based reward models by developing causally robust reward models which learn to focus on causal features for rating LLM responses [2].

Mila/Reasoning and Learning Lab *Jan 2017- Jan 2020, Sept 2023 - Present*
Research Assistant

- Several research projects on Reinforcement learning, Optimization, Representation learning and LLM Alignment [3]–[6].

Becker Friedman Institute for Economics/ Department of Economics, NYU *Jun 2020 - Sept 2023*
External Research Associate (Vendor Contract)

- Supervisor(s): Elena Manresa (NYU), Guillaume Pouliot (University of Chicago)

[1] based on the international degree equivalency information provided here

- Developed adversarial inference methods for structural estimation of econometric models. Validated its statistical properties through extensive experimentation on models with and without covariates across varied dimensions.
- Demonstrated scalability and statistical efficiency of the method in estimating high-dimensional structural models, particularly where analytical maximum likelihood estimators are unavailable. Presented findings at the American Causal Inference Conference 2022 [7].

Shared Reality Lab, McGill University *June 2021 - May 2022*
Machine Learning lead

- Led a team of software engineers to deploy advanced ML models enhancing internet accessibility for visually impaired users. Improved scalability and inference time of ML models by designing Docker-based containerization strategies hosted within a micro-services architecture.
- Collaborated with cross-functional teams (Audio, Haptic, UI-UX) to optimize ML models for rendering auditory, tactile, and visual content customized to user needs. For more information, please visit https://image.a11y.mcgill.ca/index.htm.

Microsoft Research *Nov 2017 - Jun 2018*
Research Intern

- Designed multi-modal Deep-RL agents with CNN and LSTM/GRU sequence-to-sequence models to generate natural language responses to image-based questions.
- Developed a novel implicit density estimation algorithm, integrating off-policy learning, inverse RL, energy-based models, and adversarial inference to enable human-like response generation. Assessed the generalization and grounding capabilities of multi-modal Deep-RL agents using the proposed density estimation method [3].

Fiat Chrysler Automobiles (KPIT Technologies) *Feb 2014 - Jul 2016*
Software/Controls Engineer

- Integrated neural network models into real-time engine control systems for improving fuel efficiency and reduce sensor dependency. Implemented and tested second-order optimization methods for training neural network models.
- Collaborated with relevant stakeholders to gather software requirements and ensure their accurate implementation. Developed in-vehicle and hardware-in-loop (HIL) testing strategies.

SKILLS

Programing Languages:	Python, C
Packages & Software Tools:	Python Scientific and Deep Learning Stack.
Expertise:	Stochastic Optimisation, Deep Reinforcement Learning, Large Language Models, Bayesian Experiment Design.

PUBLICATIONS

[1] **Gandharv Patil**, K. Tang, R. Aoki, and L. Guelman, *Causal epinets: Precision-corrected bounds on individual treatment effects using epistemic neural networks*, 2026. arXiv: 2605.07065 [stat.ML]. [Online]. Available: https://arxiv.org/abs/2605.07065.

[2] Shrivastava, Singh, **Gandharv Patil, *et al.***, "Robust reward modeling via causal rubrics," in *The Fourteenth International Conference on Learning Representations ICLR)*, 2026. eprint: 2506.16507. [Online]. Available: https://arxiv.org/abs/2506.16507.

[3] Gandharv Patil, "Min-max inverse reinforcement learning for learning bi-modal dialogue policies," Master's Thesis, McGill University, 2020. [Online]. Available: https://srl.mcgill.ca/publications/thesis/2020-MASTER-Patil.pdf.

[4] **Gandharv Patil**, L. Prashanth, D. Nagaraj, and D. Precup, "Finite time analysis of temporal difference learning with linear function approximation: Tail averaging and regularisation," in *Twenty-Sixth International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023. DOI: 10.48550/ARXIV.2210.05918.

[5] A. Jain, **Gandharv Patil**, A. Jain, K. Khetarpal, and D. Precup, "Variance penalized on-policy and off-policy actor-critic," in *Thirty-Fifth AAAI Conference on Artificial Intelligence*, 2021. DOI: https://doi.org/10.1609/aaai.v35i9.16964.

[6] **Gandharv Patil**, A. Mahajan, and D. Precup, "On learning history-based policies for controlling markov decision processes," in *Twenty-Seventh International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024. DOI: https://doi.org/10.48550/arXiv.2211.03011.

[7] **Gandharv Patil** and E. Manresa, "On the effectiveness of adversarial estiamtion for indirect estimation," in *American Conference on Causal Inference - ACIC*, 2022.